

HTML

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What are We Talking About Today?

- **Introduction to HTML**

- HTML: Markup language for web content
- Evolution: From Tim Berners-Lee to HTML5

- **HTML Structure**

- Elements & Tags: `<p>`, `<a>`, `<img>`, etc.
- Attributes: Define element properties (e.g., `src`, `href`)

- **Advanced HTML Features**

- Forms, Tables, Multimedia: Enhanced content
- Validation: Ensuring code quality with tools like W3C Validator

What is HTML?

- HTML stands for **H**ypertext **M**arkup **L**anguage
 - It is a **markup language**, which means it is used for processing, definition, and display of data
- HTML is a tag-based language, similar to XML
 - HTML tags are already defined though

Version	Year
HTML	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
HTML5	2014

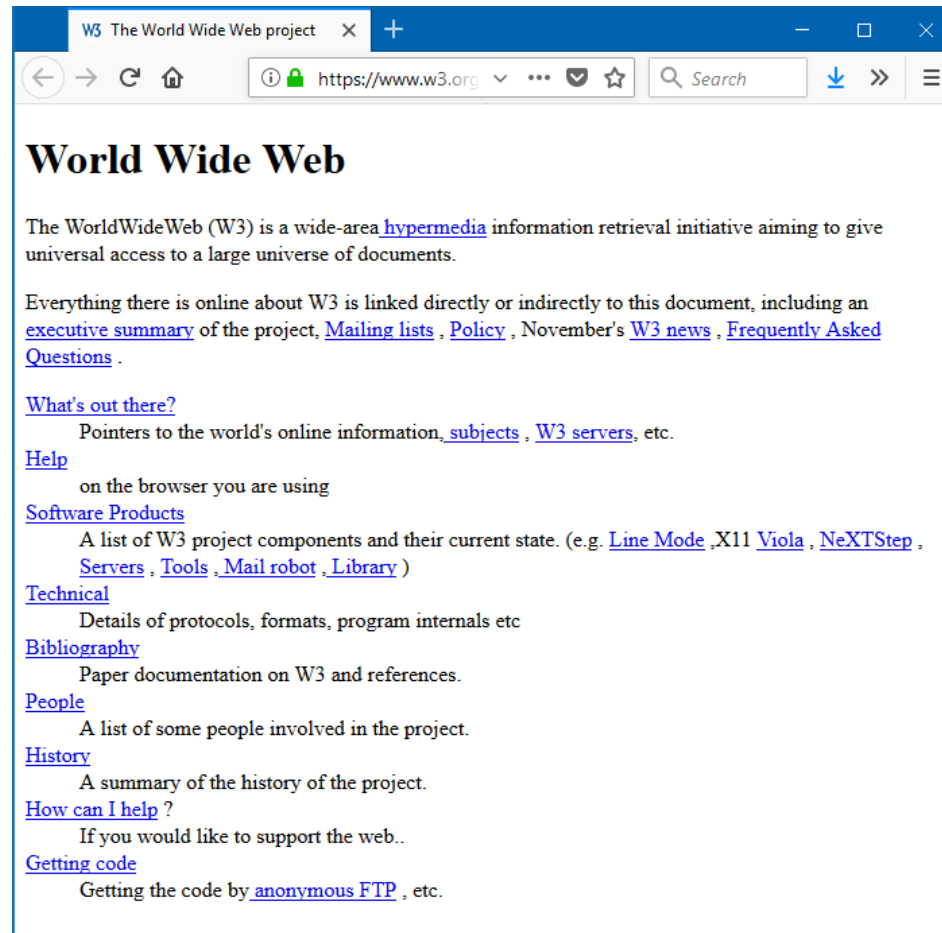


What is HTML? (cont'd)

- Hypertext Markup Language (HTML) is a language in which one can describe:
 - The display and format of text
 - The display of graphics
 - Pointers to other html files
 - Pointers to files containing graphics, digitized video and sound
 - Forms that capture information from the viewer
- HTML was developed by **Tim Berners-Lee** of CERN around 1990
- HTML is supported by Web browsers (e.g., Internet Explorer, Firefox, Chrome, Safari, many others) as well as other applications (such as mail clients, etc) – which interpret and display the output to the viewer

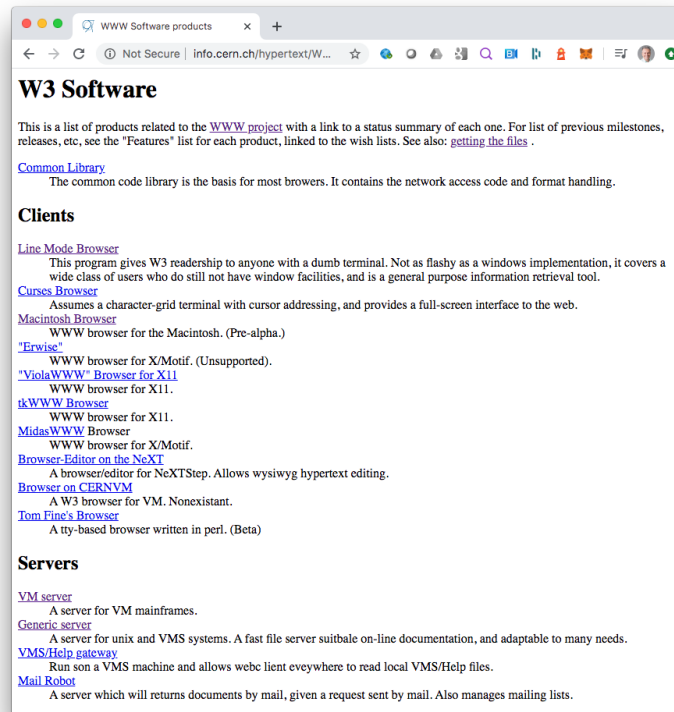
First Web Page

- The first web page ever displayed by Tim Berners-Lee at CERN



First Web Site

- The original web site of the “WWW Project” is still available at CERN
- <http://info.cern.ch/hypertext/WWW/TheProject.html>
- Includes documentation on the original browsers and servers



World Wide Web Motivation

- Originally created in 1989 at **CERN** (the European Organization for Nuclear Research) for sharing research in a faster manner than journal publications allowed

930430

ORGANISATION EUROPEENNE POUR LA RECHERCHE NUCLEAIRE
CERN EUROPEAN ORGANIZATION FOR NUCLEAR RESEARCH

STATEMENT CONCERNING CERN W3 SOFTWARE RELEASE INTO PUBLIC
DOMAIN

TO WHOM IT MAY CONCERN

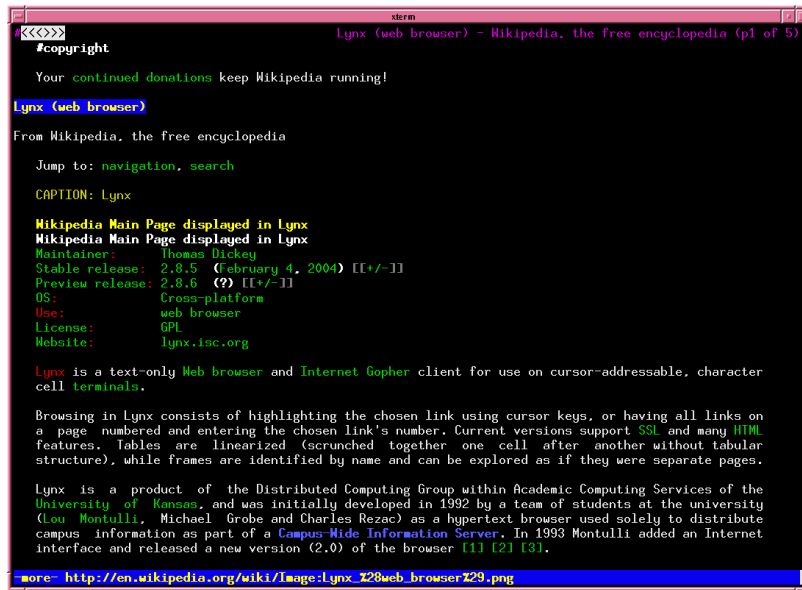
Introduction

The World Wide Web, hereafter referred to as W3, is a global computer networked information system.

The W3 project provides a collaborative information system independent of hardware and software platform, and physical location. The project spans technical design notes, documentation, news, discussion, educational material, personal notes, publicity, bulletin boards, live status information and numerical data as a uniform continuum, seamlessly intergated with similar information in other disciplines.

Text Browser – Lynx

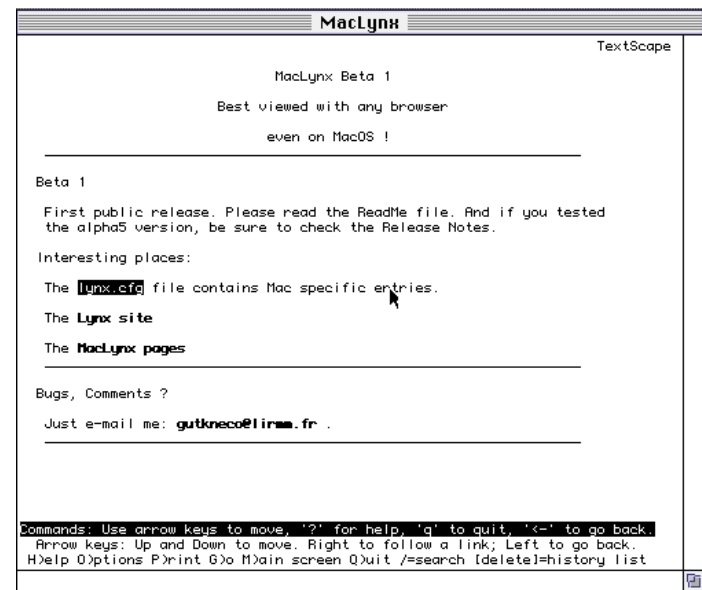
- Developed by Distributed Computing Group within Academic Computing Services of the University of Kansas
- Team of students and staff with initial release in 1992
- [https://en.wikipedia.org/wiki/Lynx_\(web_browser\)](https://en.wikipedia.org/wiki/Lynx_(web_browser))



A screenshot of the Lynx web browser running in a terminal window. The title bar reads "Lynx (web browser) - Wikipedia, the free encyclopedia (p1 of 5)". The content shows the Wikipedia main page with a copyright notice, a link to navigation, and a list of release information. The browser status bar at the bottom shows the URL: http://en.wikipedia.org/wiki/Image:Lynx_22web_browser229.png.

```

Lynx (web browser) - Wikipedia, the free encyclopedia (p1 of 5)
#copyright
Your continued donations keep Wikipedia running!
Lynx (web browser)
From Wikipedia, the free encyclopedia
Jump to: navigation, search
CAPTION: Lynx
Wikipedia Main Page displayed in Lynx
Wikipedia Main Page displayed in Lynx
Maintainer: Thomas Dickey
Stable release: 2.8.5 (February 4, 2004) [[+/-]]
Preview release: 2.8.6 (?) [[+/-]]
OS: Cross-platform
Use: web browser
License: GPL
Website: lynx.isc.org
Lynx is a text-only Web browser and Internet Gopher client for use on cursor-addressable, character cell terminals.
Browsing in Lynx consists of highlighting the chosen link using cursor keys, or having all links on a page numbered and entering the chosen link's number. Current versions support SSL and many HTML features. Tables are linearized (scrunched together one cell after another without tabular structure), while frames are identified by name and can be explored as if they were separate pages.
Lynx is a product of the Distributed Computing Group within Academic Computing Services of the University of Kansas, and was initially developed in 1992 by a team of students at the university (Lou Montulli, Michael Grobe and Charles Rezac) as a hypertext browser used solely to distribute campus information as part of a Campus-Wide Information Server. In 1993 Montulli added an Internet interface and released a new version (2.0) of the browser [1] [2] [3].
more: http://en.wikipedia.org/wiki/Image:Lynx_22web_browser229.png
```



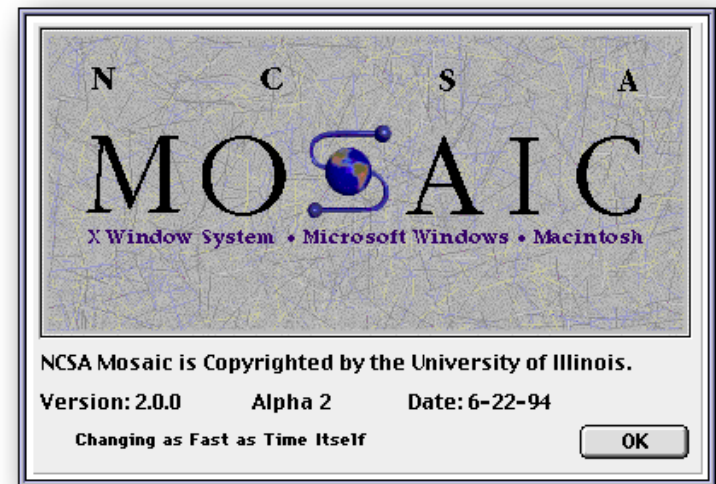
A screenshot of the MacLynx web browser running in a TextScape window. The title bar reads "MacLynx". The content displays the MacLynx Beta 1 page, which includes a welcome message, release information, and a list of interesting places. The browser status bar at the bottom shows the URL: http://en.wikipedia.org/wiki/Image:Lynx_22web_browser229.png.

```

MacLynx
TextScape
MacLynx Beta 1
Best viewed with any browser
even on MacOS !
Beta 1
First public release. Please read the README file. And if you tested the alpha5 version, be sure to check the Release Notes.
Interesting places:
The lynx.cfg file contains Mac specific entries.
The Lynx site
The MacLynx pages
Bugs, Comments ?
Just e-mail me: gutknecht@linas.fr.
Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.
Arrow keys: Up and Down to move, Right to follow a link, Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```

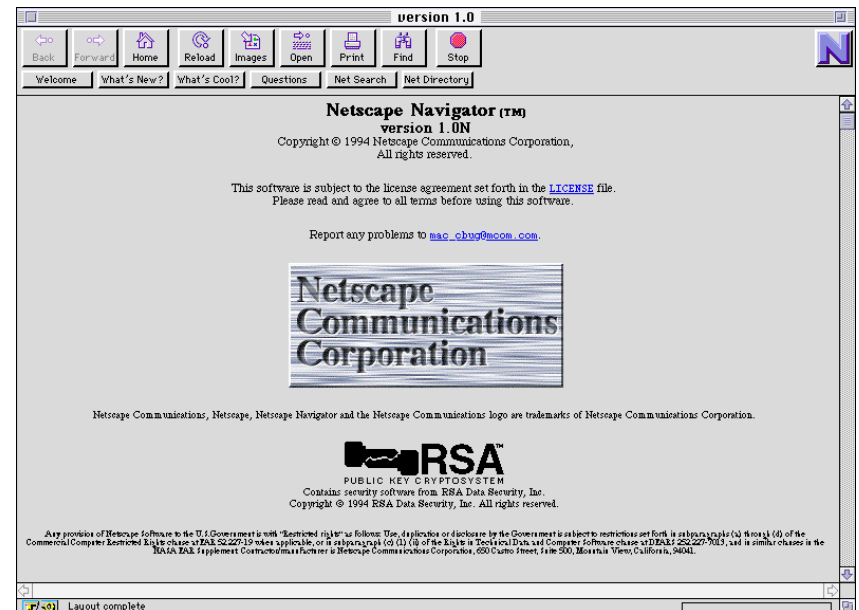
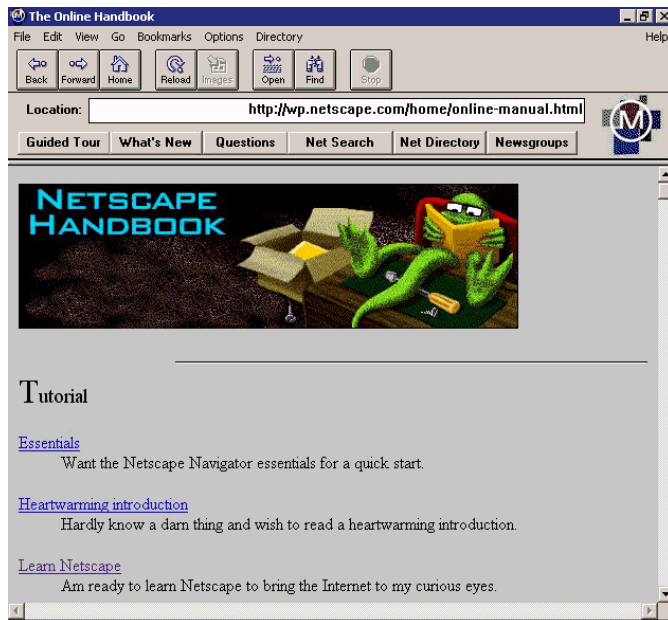

First Graphical Browser – NCSA Mosaic

- Developed at the National Center for Supercomputing Applications (NCSA) at the University of Illinois at Urbana-Champaign
- Marc Andreessen, lead student developer. NCSA released it in 1993
- <http://www.ncsa.illinois.edu/>



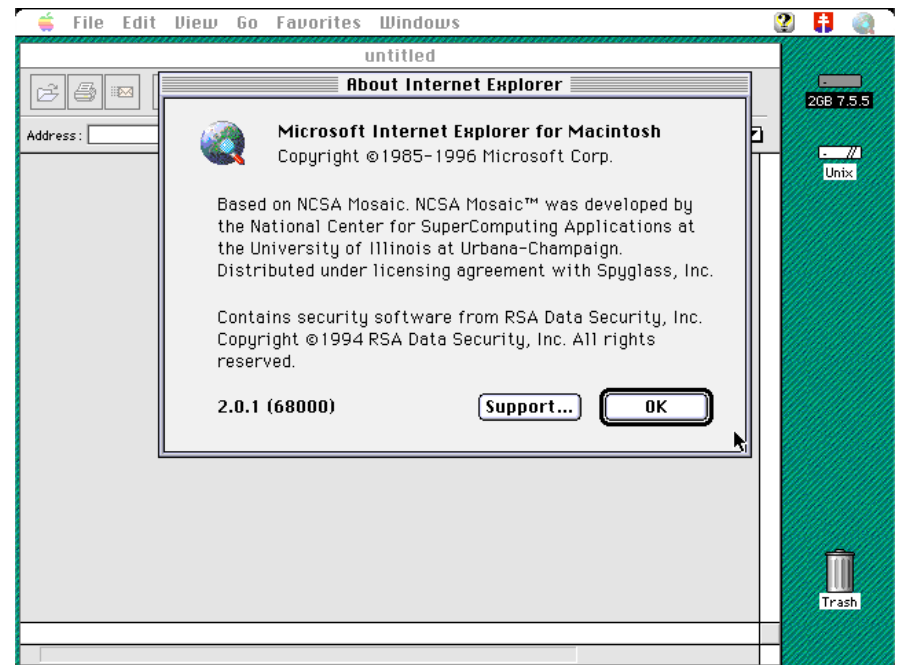
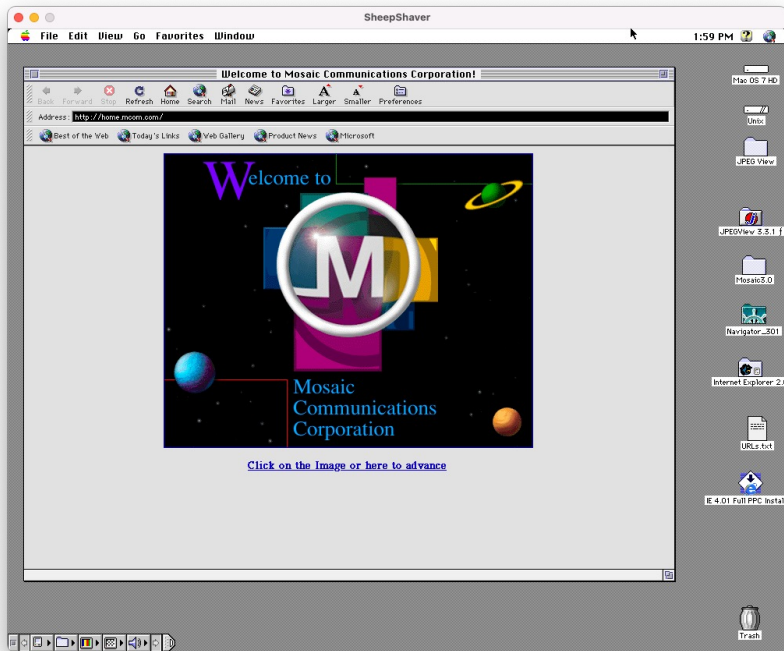
First Commercial Browser – Netscape

- Netscape **Navigator** was a proprietary web browser, and the original browser of the Netscape line
- Co-written by Marc Andreessen, with initial release in Dec. 1994
- <http://home.mcom.com>



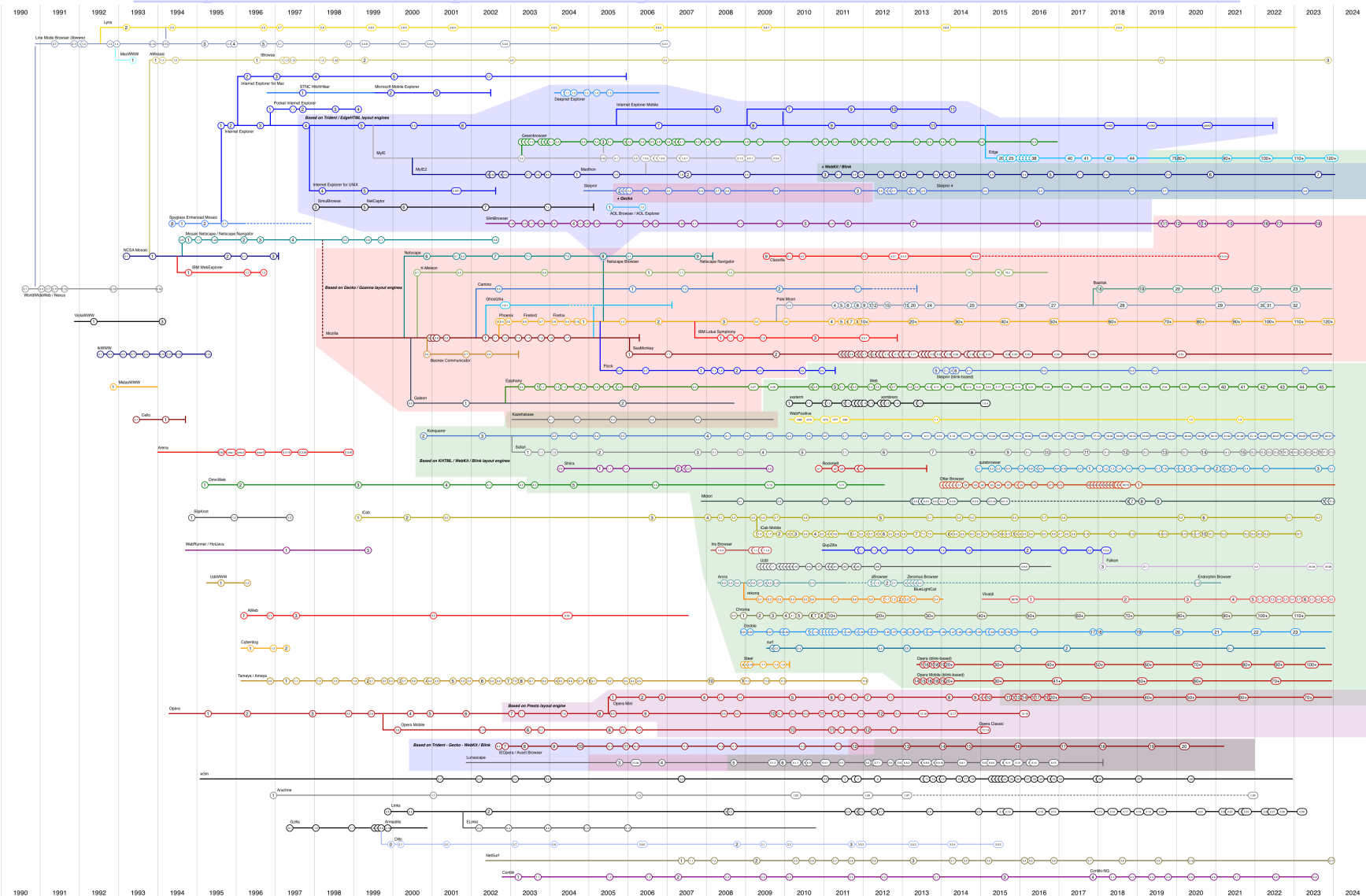
First Free Commercial Browser – IE

- Microsoft Internet Explorer (IE) killed Netscape **Navigator** as it was bundled with Windows 95
- Licensed from NCSA, with initial release in August 1995
- <https://microsoft.com/ie>

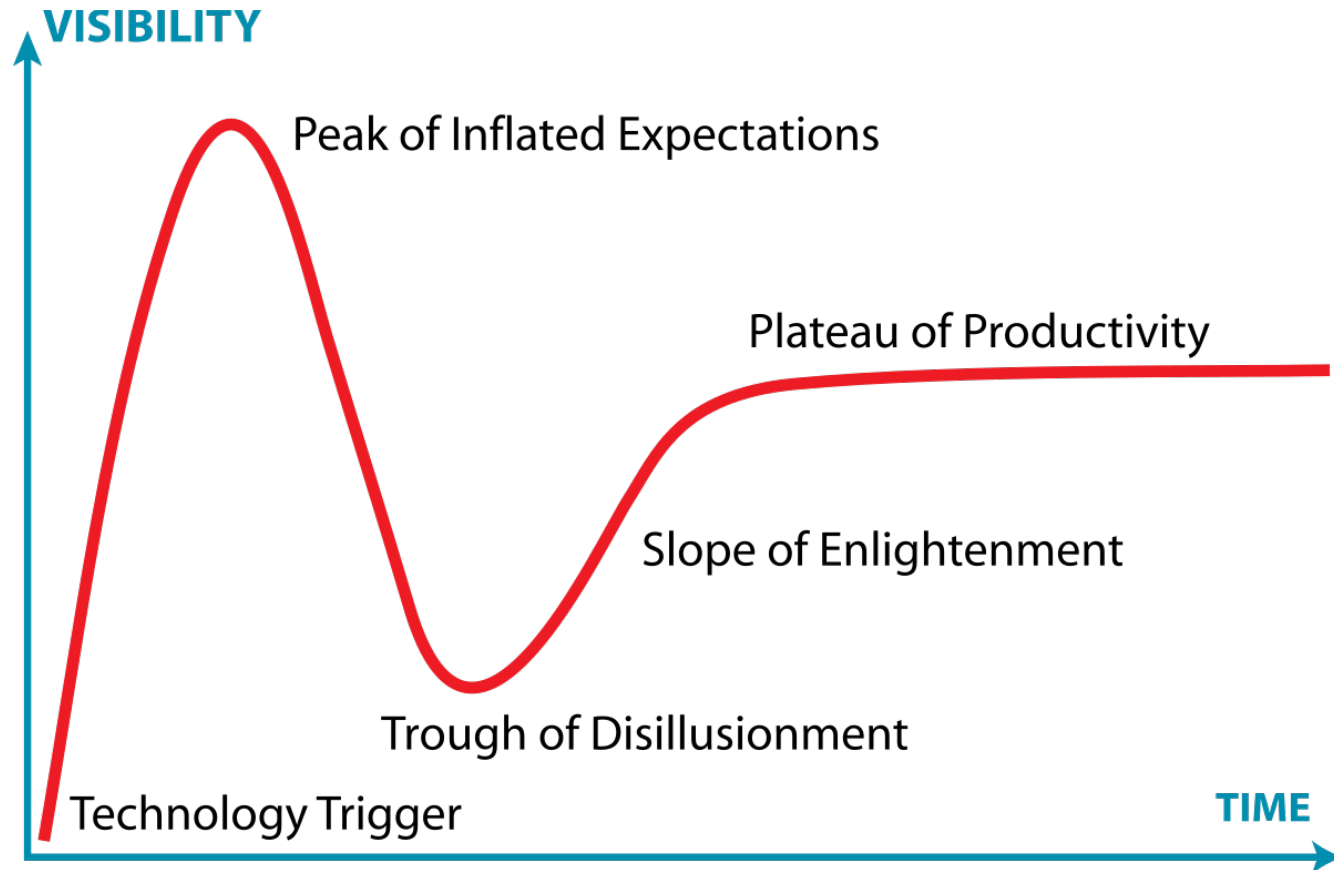


Web Browsers Timeline

http://en.wikipedia.org/wiki/List_of_web_browsers



Garner Hype Cycle



Garner Hype Cycle (cont'd)

- **Innovation Trigger**

- A potential technology breakthrough kicks things off. Early proof-of-concept stories and media interest trigger significant publicity. Often no usable products exist and commercial viability is unproven.

- **Peak of Inflated Expectations**

- Early publicity produces a number of success stories – often accompanied by scores of failures. Some companies take action; many do not.

- **Trough of Disillusionment**

- Interest wanes as experiments and implementations fail to deliver. Producers of the technology shake out or fail. Investments continue only if the surviving providers improve their products to the satisfaction of early adopters.

Garner Hype Cycle (cont'd)

- **Slope of Enlightenment**

- More instances of how the technology can benefit the enterprise start to crystallize and become more widely understood. Second- and third-generation products appear from technology providers. More enterprises fund pilots; conservative companies remain cautious.

- **Plateau of Productivity**

- Mainstream adoption starts to take off. Criteria for assessing provider viability are more clearly defined. The technology's broad market applicability and relevance are clearly paying off.

Source:

<https://www.gartner.com/en/research/methodologies/gartner-hype-cycle>

Versions of HTML

- Version 0, 1990, was the original, minimum set of HTML
- Version 1 adds highlighting and images
- Version 2, November 1995, all V.0 and V.1, plus forms
- Version 3.2, January 1997, released by W3CW, tables
- HTML 4.01, December 1999 Recommendation:
<http://www.w3.org/TR/html401/>
- HTML5, October 2014 Recommendation, vocabulary and APIs:
<http://www.w3.org/TR/html5/>
- HTML5, December 2017 HTML 5.2 Recommendation:
<https://www.w3.org/TR/html52/>
- **HTML Living Standard, December 2019:**
<https://html.spec.whatwg.org>
- W3C and WHATWG Agreement
<https://www.w3.org/blog/news/archives/7753>
- "W3C stops independent publishing of a designated list of specifications related to HTML and DOM and instead will work to take WHATWG Review Drafts to W3C Recommendations"

NOTE: some of the examples shown here are extracted from the HTML4.0 specification. This document is copyrighted according to: "Copyright © World Wide Web Consortium. All Rights Reserved. <http://www.w3.org/Consortium/Legal/>"

Elements, Tags and Attributes

- An HTML file contains Elements, Tags and Attributes
- HTML Element
 - Basic HTML node that adds semantics
 - Can be of two types:
 - Without content: self-closing start-tag
 - With content: start tag, content, end tag
- HTML Tag
 - Composed by the name of the Element surrounded by angle brackets
 - End Tag has a slash after the opening angle bracket (required in HTML5)
- Attribute
 - Specified inside Start Tag
 - Controls element properties / behavior or provides metadata

``
self-closing tag

Diagram illustrating the structure of an HTML element with content:

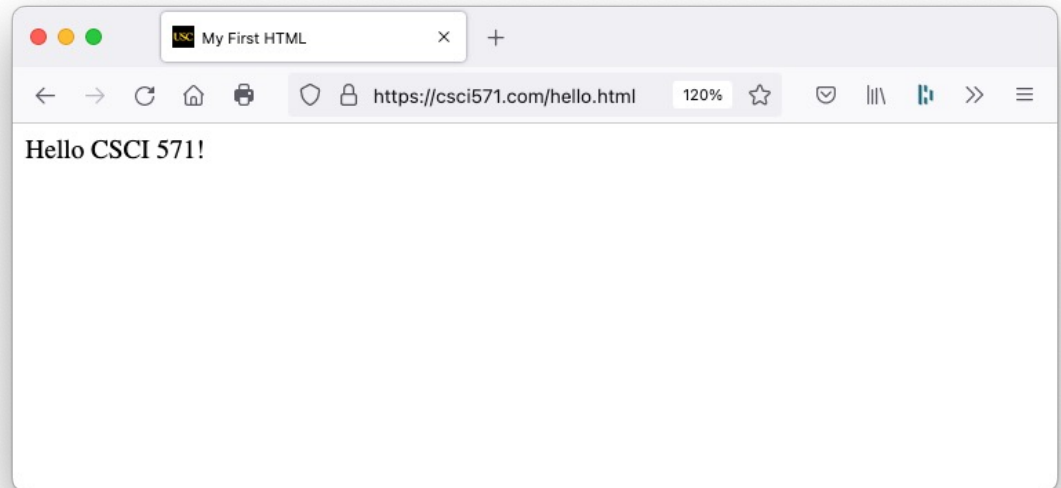
`<p class="paragraph">This is a paragraph.</p>`

The diagram labels the components of the element:

- Element**: The entire structure from `<p` to `</p>`.
- Start tag**: `<p`.
- Attribute name**: `class`.
- Attribute value**: `"paragraph"`.
- Content**: `This is a paragraph.`
- End tag**: `</p>`.

My First HTML

```
1 <!DOCTYPE html>
2 <!-- My first HTML comment -->
3 <html>
4   <head>
5     <title>My First HTML</title>
6   </head>
7   <body>
8     Hello CSCI 571!
9   </body>
10 </html>
```



General Structure

- HTML documents have a head and a body
 - *head* contains title and other **metadata**
 - *body* contains **page content** (paragraphs, etc...)
- A leading line indicates which version of HTML this document conforms to

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0//EN" "http://www.w3.org/TR/REC-html40/strict.dtd">
```

```
<HTML>
```

```
<HEAD>
```

```
<TITLE>The Solar System</TITLE>
```

```
</HEAD>
```

```
<BODY>
```

```
<P>The nine planets of the solar system are...
```

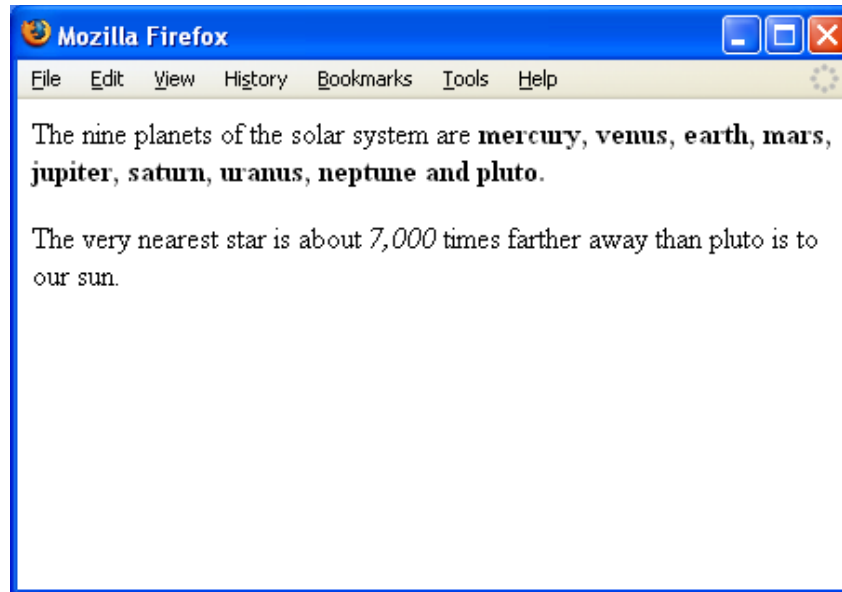
```
</BODY>
```

```
</HTML>
```

Adding Some Markup

<P>The nine planets of the solar system are mercury, venus, earth, mars, jupiter, saturn, uranus, neptune and pluto.</P>

<P>The very nearest star is about <I>7,000</I> times farther away than pluto is to our sun. </P>

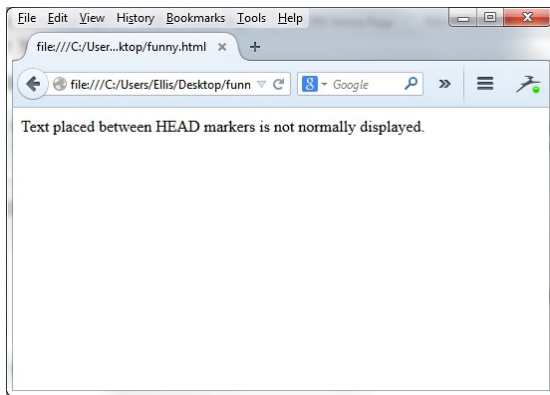


Browsers Are Tolerant

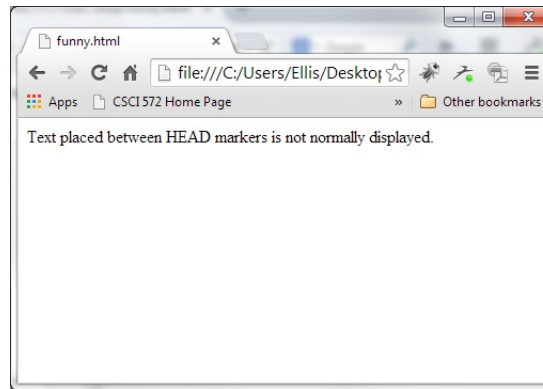
- Browsers follow the rule of being tolerant of mistakes in the input
 - They ignore markup they don't understand
- Edge, Safari, Firefox, Chrome are tolerant browsers
 - They do not insist that the HTML document begin and end with `<HTML>` and contain doctype
 - `<HEAD>` and/or `<BODY>` tags are not required
 - But there is no guarantee that this behavior will be the same for all browsers and across different versions

Browsers are Tolerant (examples)

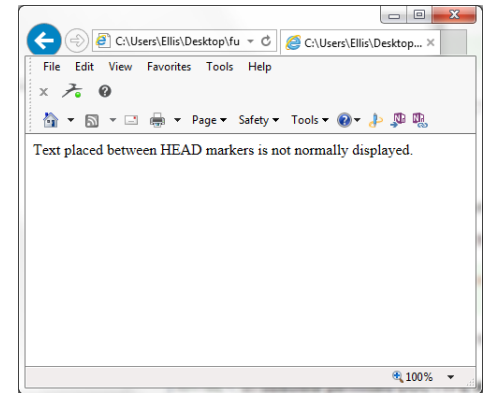
- Suppose the entire document is one line, such as:
`<HEAD>Text placed between <ODDTAG> HEAD markers is not normally displayed.</HEAD>`



Firefox



Chrome



Internet Explorer

HTML Elements

- Each element consists of:
 - a start tag,
 - content (optional),
 - an end tag (optional).
- E.g., `<BODY> some text </BODY>`
- A slash (/) after the “<” indicates an end tag
- Some elements do not require end tags, e.g.,
`<P>` paragraph tag
- Some elements do not require content, e.g.,
`<HR>`

horizontal rule tag places a straight line across the page

Attributes

- Elements may have parameters, called attributes
- Attributes are placed in the start tag before the final ">"
- Attributes have the form `name=value`

E.g., `<H1>` is the first heading tag

```
<H1 id="Chapter1">
```

```
    Start of Chapter 1
```

```
</H1>
```

- Attribute values are often enclosed in quotes, either double or single
- Quotes are not required when the value contains only letters, digits, hyphens, and periods.
- Attribute names are case insensitive,
- but attribute values are usually not

Comments in HTML

- Comments start with
 `<!--`
 and end with
 `-->`
- Comments cannot be nested
 - White space is permitted between the `--` and the closing angle bracket, `>`
 - It is not permitted between the opening angle bracket, exclamation point, and the `--`
- E.g.

`<BODY>`

`<!-- This is a comment`

`and hence not displayed -->`

But this will be displayed

`</BODY>`

Complete Set of <BODY> tag attributes

- `id`, assigns a unique name to an element

e.g., `<P id=mystart>This is my starting paragraph...`

- `class`, assigns one or more names to an element
- `lang`, a language code that identifies a natural language spoken, written, or otherwise used
- `title`, a short description of the body
- `style`, inline display information
- `bgcolor`, background color
- Events include

`onload, onunload onclick, ondblclick, onmousedown, onmouseup, onmouseover, onmousemove, onmouseout, onkeypress onkeydown, onkeyup`

- Deprecated elements in include:

`background, text, link, vlink, alink`

See <http://www.w3.org/TR/html4/struct/global.html#edef-BODY>

<BODY> Using Style Sheets

- Deprecated example

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<HTML>
<HEAD>
  <TITLE>A study of population dynamics</TITLE>
</HEAD>
<BODY bgcolor="white" text="black" link=" red" alink="fuchsia" vlink="maroon">
... document body...
</BODY>
</HTML>
```

- Using Style Sheets

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<HTML>
<HEAD>
<TITLE>A study of population dynamics</TITLE>
<STYLE type="text/css">
BODY { background: white; color: black}
A:link { color: red } A:visited { color: maroon } A:active { color: fuchsia }
</STYLE>
</HEAD>
<BODY>
... document body...
</BODY>
</HTML>
```

Composing HTML

- Conventional editors let you compose HTML directly
 - e.g., TextEdit, emacs, vi, NotePad, TextPad, etc.
 - use the tools when you are writing HTML directly
- Word Processors include a File SaveAs option which saves your document in HTML format
 - e.g., Microsoft Word
 - **Do not use!**
- There are several free HTML-specific editors, e.g.
 - **Brackets**, <http://brackets.io>
 - **Visual Studio Code**, <https://code.visualstudio.com/>
- There are several commercial HTML/CSS suites, e.g.
 - JetBrains WebStorm
 - Adobe Dreamweaver CC (Creative Cloud subscription)
- For a complete list see http://en.wikipedia.org/wiki/List_of_HTML_editors

HTML Headings

- Heading tags should be used for different levels in a document
 - The size and look can be changed with CSS, so don't use these tags solely to change the size of the font

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>My First Headings</title>
```

```
</head>
```

```
<body>
```

```
<h1>Hello CSCI 571!</h1>
```

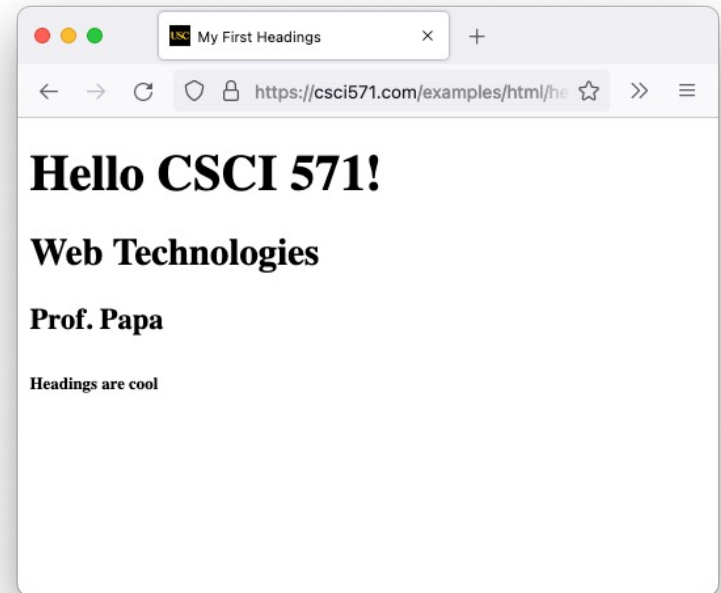
```
<h2>Web Technologies</h2>
```

```
<h3>Prof. Papa</h3>
```

```
<h6>Headings are cool</h6>
```

```
</body>
```

```
</html>
```



HTML Line Breaks

- Paragraph tags separate logical display blocks
- Line breaks are used for formatting

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>My First Paragraph</title>
```

```
  </head>
```

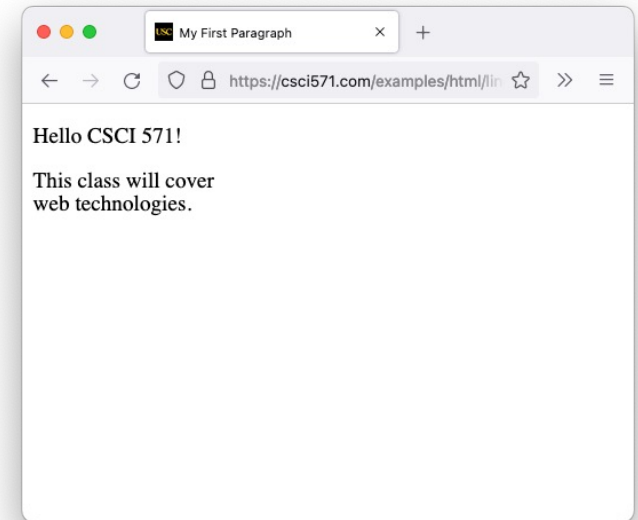
```
  <body>
```

```
    <p>Hello CSCI 571!</p>
```

```
    <p>This class will cover<b> />web technologies.</p>
```

```
  </body>
```

```
</html>
```



HTML Lists

- HTML lists are very common in everyday web development.
- Perhaps the most used are
 - unordered lists (`<ul>`) and
 - ordered lists (`<ol>`)
- but there are a few other list options
 - definition list (`<dl>`), and
 - the menu (`<menu>`) element,
 - both were deprecated in HTML4, but reintroduced in HTML5
- All lists follow the same pattern:
 - *<start tag of list>*
 - *<list item tag>*
 - *<list item tag>*
 - *<list item tag>*
 - *</ end tag of list>*

HTML Definition Lists

<DL>

 <DT>light year<DD>the distance light travels in
one year

<DT>asteroids<DD>are small, irregular shaped
objects, mostly occurring between Mars and
Jupiter

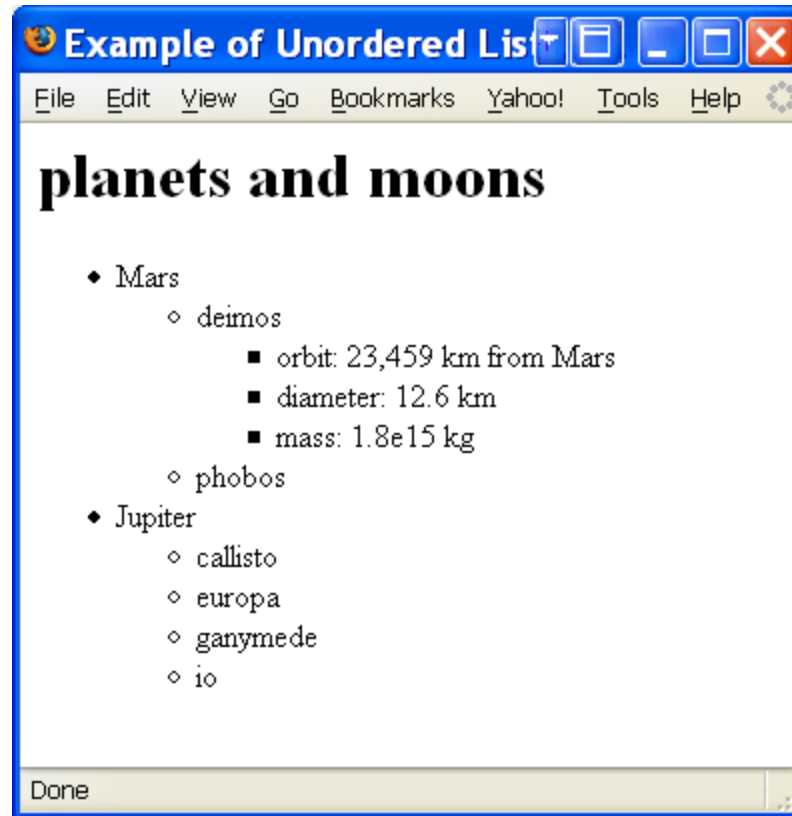
</DL>



HTML Unordered Lists

```
<HTML>
<HEAD> <TITLE>Example of Unordered Lists</TITLE> </HEAD>
<BODY>
  <H1>planets and moons</H1>
  <UL>
    <LI>Mars
      <UL>
        <LI> deimos
          <UL>
            <LI>orbit: 23,459 km from Mars
            <LI>diameter: 12.6 km
            <LI>mass: 1.8e15 kg
          </UL>
        <LI>phobos
      </UL>
    <LI>Jupiter
      <UL><LI>callisto<LI>europa<LI>ganymede<LI>io</UL>
  </UL>
</BODY></HTML>
```

Browser Output of Unordered Lists



HTML Ordered Lists

- Has the general form

`<OL><LI> first list item<LI> second list item</OL>`

- START attribute can initialize the sequence to a number other than 1
- TYPE attribute can be used to select the numbering style

Type	Name	Style
1	arabic	1, 2, 3, ...
a	lower alpha	a, b, c, ...
A	upper alpha	A, B, C, ...
i	lower roman	i, ii, iii
I	upper roman	I, II, III,

- However, the type attribute is **deprecated**, and list styles should be handled through style sheets, e.g.

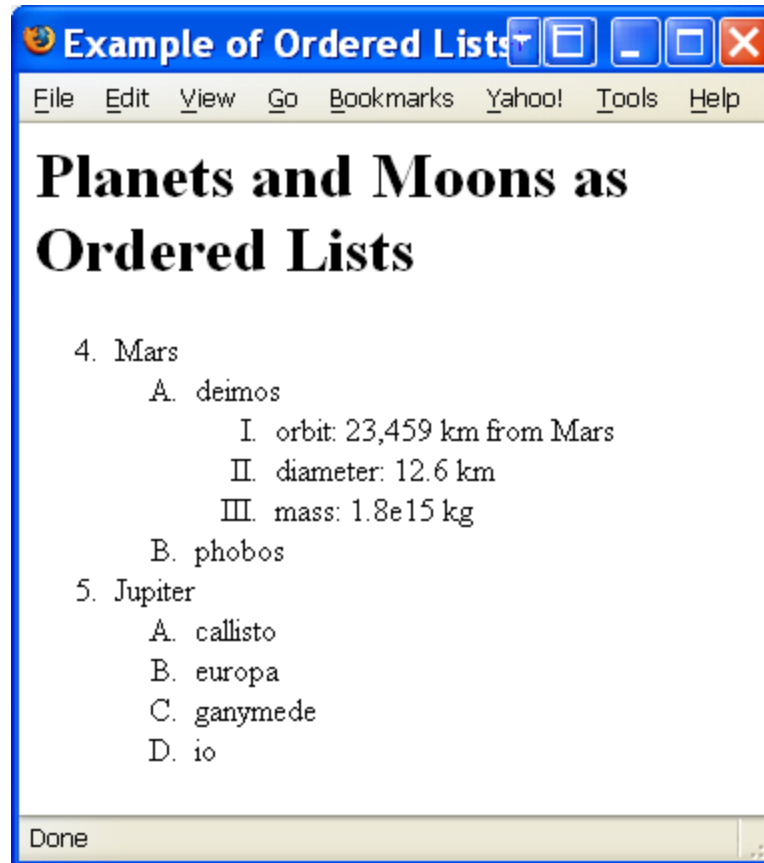
`<STYLE type="text/css">`

`OL.withroman {list-style-type: lower-roman}</STYLE>`

Example - Ordered Lists

```
<HTML><HEAD><TITLE>
Example of Ordered Lists</TITLE></HEAD>
<BODY>
  <H1>Planets and Moons as Ordered Lists</H1>
  <OL START=4>
    <LI>Mars
      <OL type=A>
        <LI>deimos
          <OL type=I>
            <LI>orbit: 23,459 km from Mars
            <LI>diameter: 12.6 km
            <LI>mass: 1.8e15 kg
          </OL>
        <LI>phobos
      </OL>
    <LI>Jupiter
      <OL type=A><LI>callisto<LI>europa<LI>ganymede<LI>io</OL>
  </OL>
</BODY></HTML>
```

Browser Output



HTML Table Elements

- `<TABLE>`, a tag used to define a table
- `<CAPTION>`, a tag to label a table
 - Its attributes are `ALIGN = top, bottom, left, right`
- `<TH></TH>` or `<TD></TD>`, tags that identify a table header cell and table data cell
 - Headers are the same as data except they use bold font and are centered
- `<TR>`, a tag that identifies a container for a row of table cells
 - Same attributes as TH and TD

Facts about Tables

- Table cells can be text, lists, images, forms, figures, or even tables
- Table headers are typically rendered in boldface, and table data is typically rendered in the regular font and point size
- A table has an optional caption followed by rows
- Table rows are said to contain table headers and table data
- The browser will set the number of columns to be the greatest number of columns in all of the rows
- Blank cells are used to fill extra columns in the rows

Example - 3 rows x 2 cols

```
<HTML>
  <HEAD> <TITLE>Table: 3 rows 2 Cols</TITLE> </HEAD>

  <BODY>
    <H1>Table: 3 Rows 2 Cols</H1>
    <TABLE BORDER="2">
      <CAPTION>MIME Content-Types</CAPTION>
      <TR>
        <TD>application/postscript</TD>
        <TD>Postscript</TD>
      <TR>
        <TD>application/rtf</TD>
        <TD>MS Rich Text Format</TD>
      <TR>
        <TD>application/x-pdf</TD>
        <TD>Adobe Acrobat Format</TD>
      </TABLE>
    </BODY>
  </HTML>
```


Browser Output

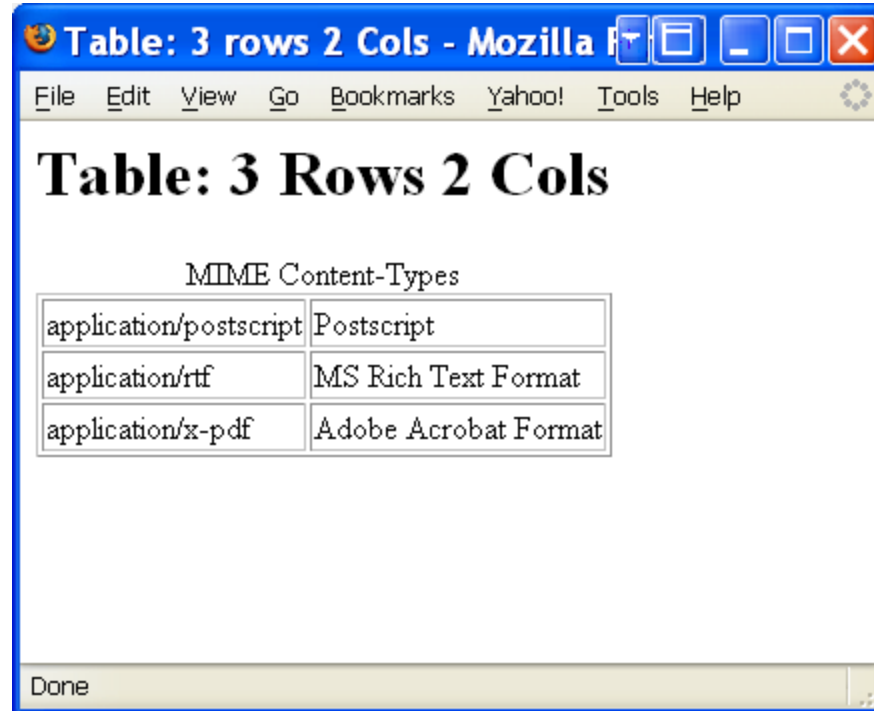
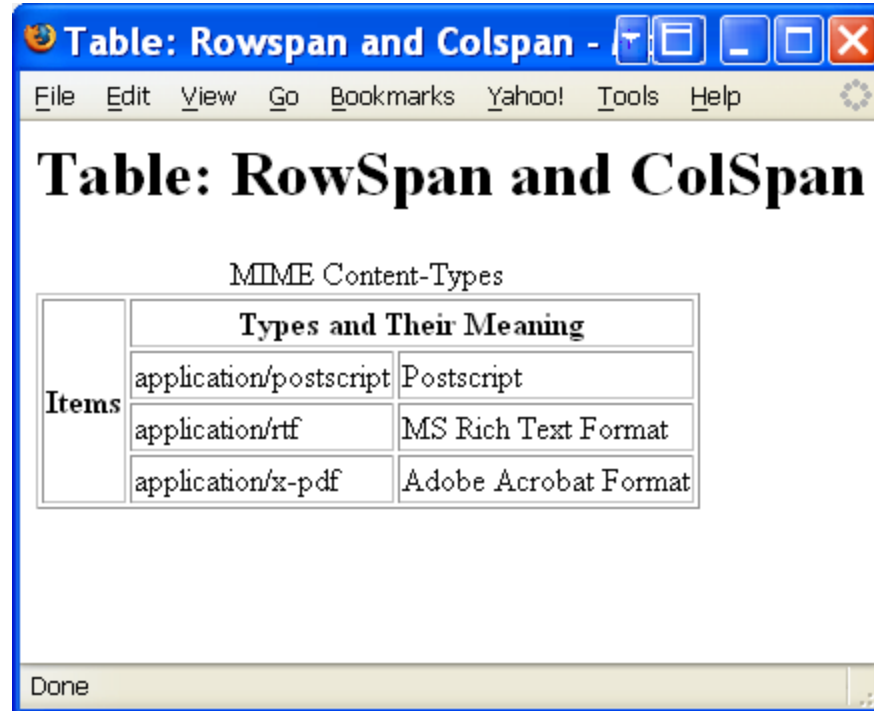


Table Example Rowspan colspan

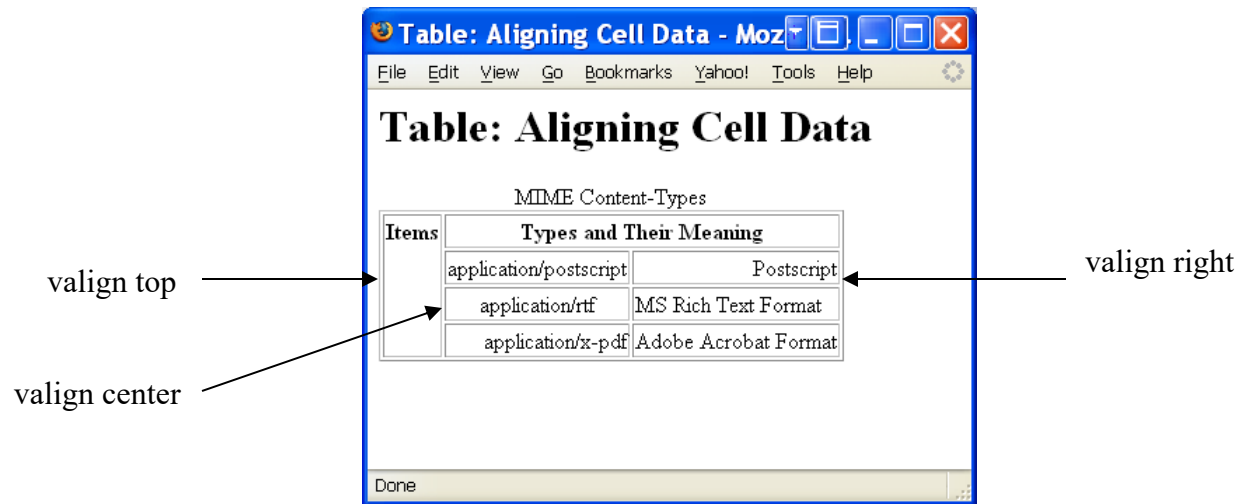
```
<HTML>
  <HEAD><TITLE>Table: Rowspan and Colspan</TITLE></HEAD>
  <BODY>
    <H1>Table: RowSpan and ColSpan</H1>
    <TABLE BORDER="2">
      <CAPTION>MIME Content-Types</CAPTION>
      <TR>
        <TH ROWSPAN=4>Items</TH>
        <TH colspan=2>Types and Their Meaning</TH>
      <TR>
        <TD>application/postscript</TD> <TD>Postscript</TD>
      <TR>
        <TD>application/rtf</TD> <TD>MS Rich Text Format</TD>
      <TR>
        <TD>application/x-pdf</TD> <TD>Adobe Acrobat Format</TD>
      </TABLE>
    </BODY>
  </HTML>
```

Browser Output



Arranging Data in a Table

- Originally data in a table could be manipulated using attributes: align left, align right, align center, valign top, valign middle and valign bottom



- The above attributes are **deprecated** in **HTML5** in favor of Cascading Style Sheets (CSS) settings
- What does "deprecated" mean?
- See the slides in the lecture on CSS

HTML Character Set

- HTML uses the **Universal Character Set (UCS)**, defined in [ISO10646]. This standard defines a repertoire of thousands of characters used by communities all over the world.
 - Its latest specification (**Unicode 16.0.0**), dated September 10, 2024, can be found at <http://www.unicode.org/versions/latest/>
 - Includes Unicode Emoji (with skin tone diversity)
- HTML must also specify how characters are encoded during transmission.
- Commonly used character encodings on the Web include
 - ISO-8859-1 (also referred to as "Latin-1",
 - ISO-8859-5 (which supports Cyrillic),
- A browser is informed of the encoding by a line
Content-Type: text/html; charset=EUC-JP

Character references

- Character references in HTML may appear in two forms:
 - **Numeric** character references (either decimal or hexadecimal)
 - `å` (in decimal) represents the letter "a" with a small circle above it (used, for example, in Norwegian).
 - `<` represents left angle bracket
 - `>` represents right angle bracket
 - `&` represents ampersand sign
 - `"` represents double quote
 - Character **entity** references.
 - `"<"` represents the `<` sign.
 - `">"` represents the `>` sign.
 - `"&"` represents the `&` sign.
 - `"""` represents the `"` mark.

Example - Character References

<HTML>

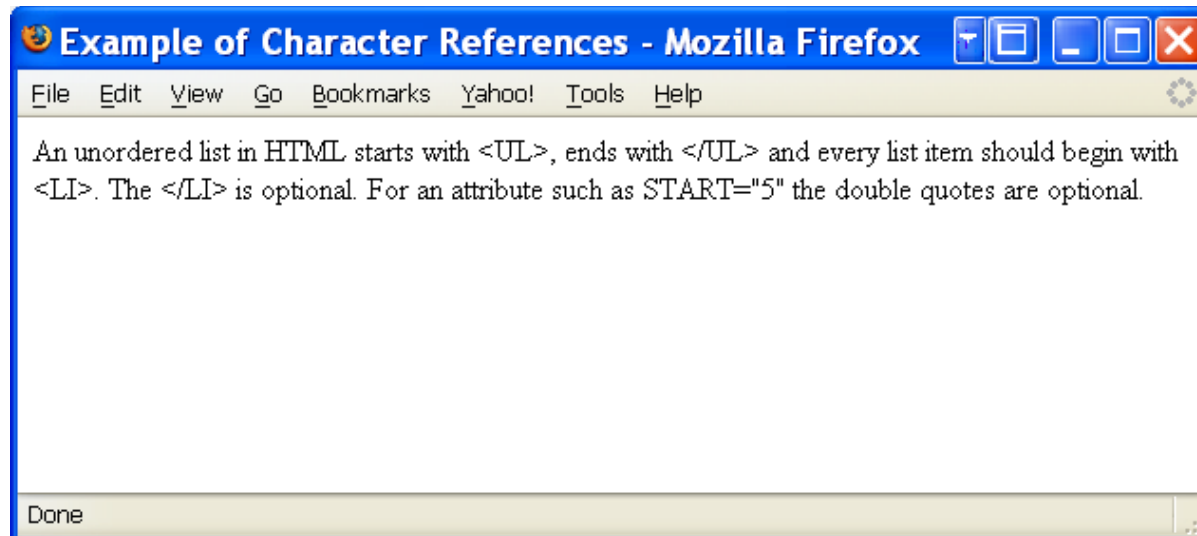
<HEAD><TITLE>Example of Character References</TITLE></HEAD>

<BODY>

An unordered list in HTML starts with `<ul>`, ends with `</ul>` and every list item should begin with `<li>`. The `</li>` is optional. For an attribute such as `START="5"` the double quotes are optional.

</BODY>

</HTML>



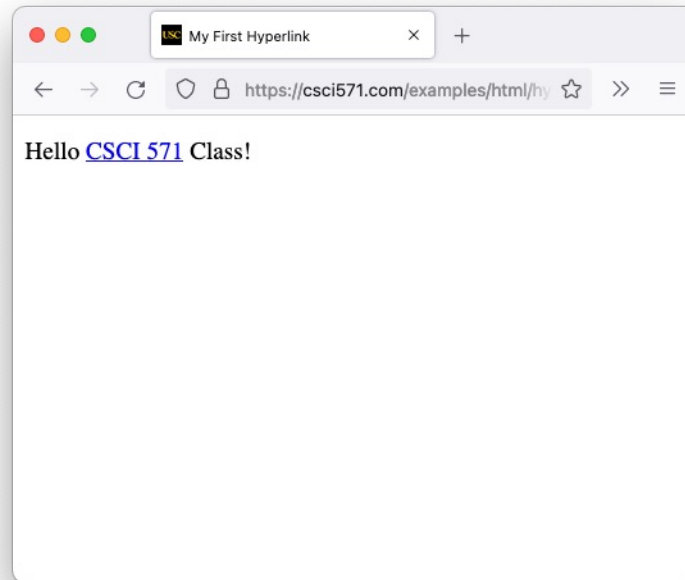
HTML Hyperlinks (Anchors)

- Hyperlinks are the “novel idea” introduced by Tim Berners-Lee
- Implemented in HTML with Anchor Tags
- An anchor is a way to designate a **link to another document** or to a specific place in the **same document**
- Begins with <A> and ends with
- The link location is given by the “required” **HREF attribute** (Hypertext REFerence); e.g.,
`<A HREF="http://csci571.com/index.html">Class Home Page</A>`
- Hypertext links are displayed using underlining, color, and/or highlighting
 - Depends on the browser defaults or style settings
 - Once a link is taken, it should change color
 - HREF, stands for **Hypertext REFerence**
- Link destination can be relative or absolute

HTML Hyperlinks (cont'd)

- Anchor tags allow one page to link to another page

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Hyperlink</title>
  </head>
  <body>
    <p>Hello <a href="https://csci571.com">CSCI 571</a> Class!</p>
  </body>
</html>
```



Syntax of Anchor Names

- An anchor name is the value of either the name or id attribute when used in the context of anchors.
- Anchor names must observe the following rules:
 - **Uniqueness:** Anchor names must be unique within a document. Anchor names that differ only in case may not appear in the same document.
 - **String matching:** Comparisons between fragment identifiers and anchor names must be done by exact (case-sensitive) match.
- See <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/a>

Defining Anchors Using the id Attribute

- The id attribute may be used to create an anchor at the start tag of any element (including the A element).
- Example: the **id attribute** places an anchor in an H2 element.

You may read more about this in

```
<A href="#section2">Section Two</A>.
```

```
. . . more text . . .
```

```
<H2 id="section2">Section Two</H2>
```

```
. . . more text
```

```
<P>Please refer to <A href="#section2">Section Two</A> above for  
more details.
```

- The id and name attributes share the same name space. They cannot both define an anchor with the same name in the same document. Try this example in IE and Firefox:

```
<P><A href="#label1">...</A>
```

```
...more document...
```

```
<P><H2 id="LABEL1">...</A>
```

← **NOTE: browsers work differently**

Examples of Anchors

- `<A href=myfile.html>` a file in the same directory and same domain as the current page
- `<A href=http://nunki.usc.edu/mydocs/book.doc>` a file in directory mydocs on machine nunki.usc.edu, a WWW site
- `<A href=news:comp.compilers>` the newsgroup computers.compilers
- `<A href=mailto:sales@xyz.com>` opens an an e-mail window for sending a message
- `<A HREF="ftp://ds.internic.net/rfc/rfc1866.txt">` Download RFC1866 `</A>` executes the ftp program to fetch a file Given the current position, this `<HREF>` moves up one directory, connects to Docs/Style/ and displays the document Overview.html, at the document "reference" anchor.
- `<A HREF=" ../Docs/Style/Overview.html#reference">` ... `</A>` Connects to lycos and runs pursuit with three arguments
- `<A HREF="http://www.lycos.com/cgi-bin/pursuit?Hypertext+Markup+Language">` ... `</A>`

Anchor Titles

- The title attribute may be set to add information about the nature of a link.
- This information may be **spoken** by a user agent, rendered as a **tooltip**, cause a change in cursor image, etc.

<BODY>

...some text...

<P>You'll find a lot more in <A href="chapter2.html"
 title="Go to chapter two">chapter two.

See also this <A href="../../../images/solarsystem.gif"
 title="GIF image of solar system">view of the
solar system.

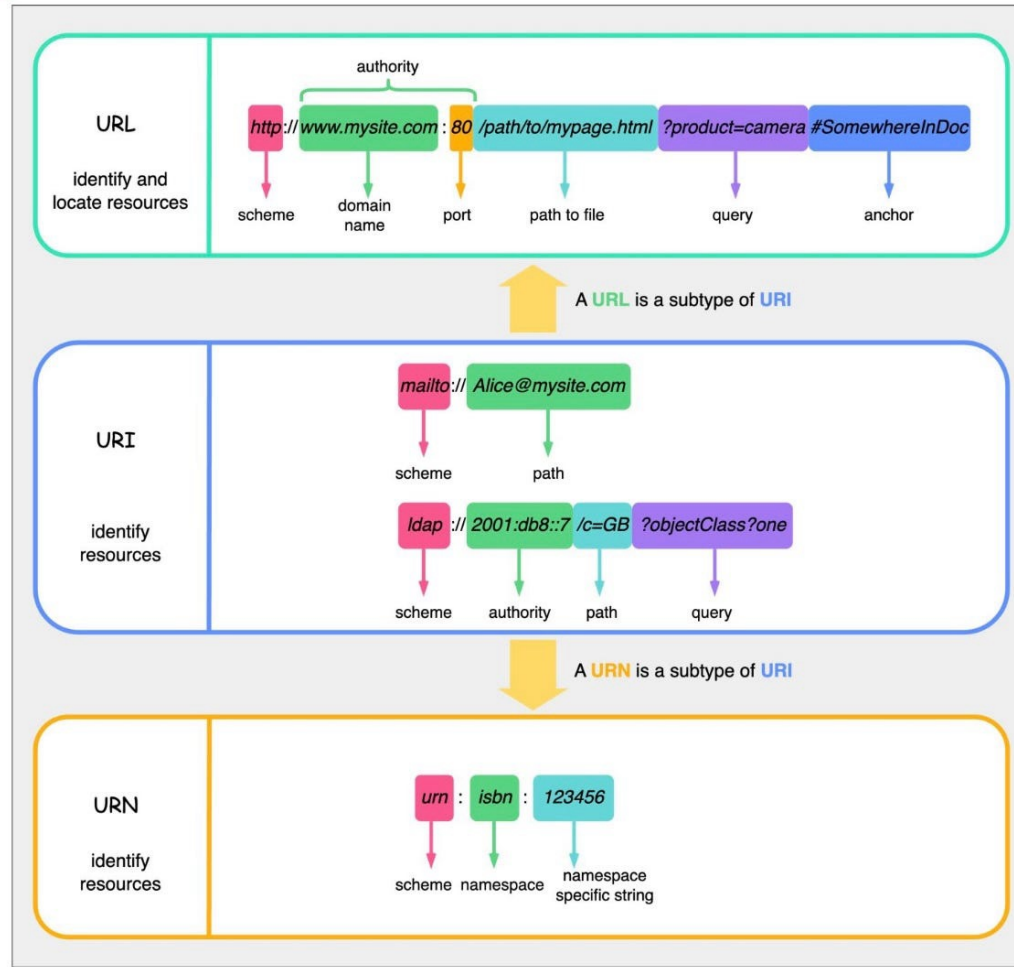
</BODY>

Universal Resource Identifier (URI)

Universal Resource Locator (URL)

URL vs URI vs URN

 blog.bytebytego.com



<https://blog.bytebytego.com/>

Universal Resource Identifier (URI) cont'd

- URIs typically consist of three pieces:
 - The **scheme** of the mechanism used to access the resource.
 - The name of the machine **hosting** the resource.
 - The name of the **resource** itself, given as a **path**E.g., `http://www.usc.edu/dept/cs/index.html`
- Fragment identifiers are URIs that refer to a location within a resource
e.g., `http://www.usc.edu/dept/cs/index.html#section2`
- Relative URIs have a path that refers to a resource on the same machine as the current document, e.g., `".."` means one level up
- See p. 24 in:
`https://datatracker.ietf.org/doc/html/rfc3986`
- See also:
`https://en.wikipedia.org/wiki/Fragment_identifier`

The <LINK> Element

- Should only appear in the HEAD
- It may appear any number of times
- It conveys relationship information that may be rendered in a variety of ways (e.g., a tool-bar with a drop-down menu of links, external files)
- Example - The current document is "Chapter2.html". The rel attribute specifies the relationship of the linked document with the current document.

```
<HTML>
```

```
<HEAD>
```

```
  <TITLE>Chapter 2</TITLE>
```

```
  <LINK rel="Index" href="../index.html">
```

```
  <LINK rel="Next" href="Chapter3.html">
```

```
  <LINK rel="Prev" href="Chapter1.html">
```

```
</HEAD>
```

```
  ...the rest of the document...
```


How is <LINK> Used

- To provide a variety of information to search engines:
 - Links to **alternate versions** of a document, written in another human language, e.g.

```
<LINK lang="fr" title="La documentation en Fran&ccedil;ais"  
type="text/html" rel="alternate" hreflang="fr"  
href="http://domain/manual/french.html">
```

- Links to alternate versions of a document, designed for **different media**

```
<LINK media="print" title="The manual in postscript"  
type="application/postscript" rel="alternate"  
href="http://domain/manual/usermanual.ps">
```

- Links to the starting page of a collection of documents.
- Links to style sheets and "**media queries**" used in Responsive Web Design

Creating Graphics

- Digital cameras & Smartphones
 - Snap and the image is digitized and can be transferred to a computer
 - Typical resolutions are 1280x720, 1920x1080
- Graphic editors
 - Permit the combination of text, drawing, and color
 - For example, Adobe Photoshop
- Scanners
 - Convert text and graphics into machine readable form

Image Formats

- Five image formats are always supported by Web browsers
 - x-pixelmaps (obsolete)
 - Similar to x-bitmaps, but 8 bits are given to each pixel, permitting 256 colors in the image
 - Graphic Interchange Format (**GIF**)
 - Support black and white, grayscale, and color
 - Patented by Unisys (expired, abandoned)
 - Joint Photographic Experts Group (**JPEG**)
 - Designed for photographic images
 - Includes image compression
 - Portable Network Graphics (**PNG**)
 - An open, extensible image format with Lossless Compression
 - Patent-free replacement for GIF and TIFF
 - W3C Recommendation: <http://www.w3.org/TR/PNG/>
 - Scalable Vector Graphic (**SVG**) - since 2008

HTML Image Element

- The **IMG** element embeds an image in the current document, e.g., `<IMG SRC="file.gif">`
- Some attributes of `<IMG>` include
 - **SRC**, the HREF or name of the image
 - **SRCSET & SIZES**, HREFs and sizes for different screen sizes and types
 - `Align=top, middle, or bottom` to align text around an image
 - **height** and **width** to control the dimensions of the image
 - `Alt` to replace an image with text, if the image is unavailable or a text browser is used; e.g.,
`<IMG SRC="eiffel.gif" ALT="picture of the Eiffel Tower">`

HTML Images (cont'd)

- There are some modern image formats supported
 - **WEBP** (2010, open format, based on VP8 codec)
 - Baseline (96% adoption - except IE or old browsers <2020)
 - **AVIF** (2019, open format, based on AV1 codec)
 - Baseline 2024 (93% adoption - supported in Edge since 2024)
 - **JPEG XL** (2021, open format)
 - Limited availability (12% adoption - Safari only, support (behind a flag) was dropped by Chrome in 2022)
 - **HEIF** - container for different formats (AVIF, HEIC, JPEG, WXAM)
 - Limited availability (12% adoption - Safari only)
- Those formats could save a lot of bandwidth; however, we need to check if they are supported: **<picture>** tag + **<source type=**



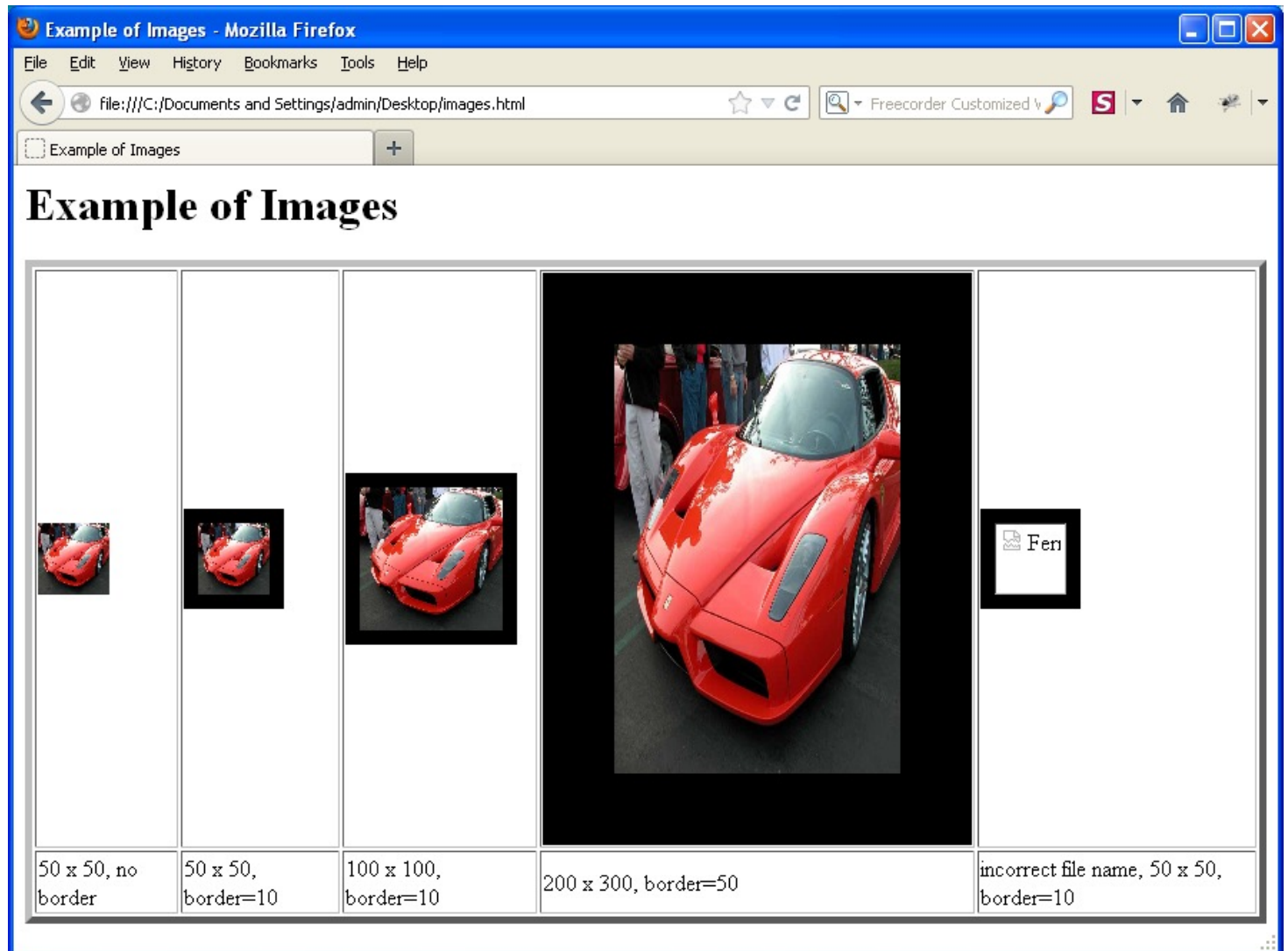
```
<picture>
  <source srcset="photo.avif" type="image/avif" />
  <source srcset="photo.webp" type="image/webp" />
  
</picture>
```

Will use avif if supported, otherwise - webp if supported, otherwise - jpg

Example - Images

```
<HTML><HEAD><TITLE>Example of Images</TITLE></HEAD>
<BODY>
<H1>Example of Images</H1>
<table border=5>
  <tr>
    <td><IMG SRC="ferrari.jpg" ALT="Ferrari" BORDER=0 WIDTH=50 HEIGHT=50>
    <td><IMG SRC="ferrari.jpg" ALT="Ferrari" BORDER=10 WIDTH=50 HEIGHT=50>
    <td><IMG SRC="ferrari.jpg" ALT="Ferrari" BORDER=10 WIDTH=100 HEIGHT=100>
    <td><IMG SRC="ferrari.jpg" ALT="Ferrari" BORDER=50 WIDTH=200 HEIGHT=300>
    <td><IMG SRC="xferrari.jpg" ALT="Ferrari" BORDER=10 WIDTH=50 HEIGHT=50>
  </tr>
  <tr>
    <td>50 x 50, no border</td>
    <td>50 x 50, border=10</td>
    <td>100 x 100, border=10</td>
    <td>200 x 300, border=50</td>
    <td>incorrect file name, 50 x 50, border=10</td>
  </tr>
</table>
</BODY></HTML>
```

Browser Output



Active Images

- Active images are images that can be clicked and, just like an anchor, they act as a hypertext link

```
<A HREF="http://sunset.usc.edu:8080/index.html">
```

```
<IMG SRC="USCimage.gif"> </A>
```

- Active images have a border around them and the cursor changes shape when passed over

Image Maps

- Image maps are active images with multiple clickable regions
- each region can be associated with a specific action (e.g., retrieve a document, run a program, etc.)
- When the region is activated by the user, e.g., by a mouse click, the action is taken
 - the pixel coordinates are interpreted by the browser (**USEMAP** attribute).
 - The **<map>** tag is used to associate the image and the regions.

An ImageMap Example from Wikipedia



To see the ImageMap work
go to

http://en.wikipedia.org/wiki/Image_map
and click on each of the individuals
at the table

```

```

```
<map id="ImageMap_1_2013620197"  
name="ImageMap_1_2013620197">
```

```
<area title="Dr Johnson – Dictionary writer"  
alt="Dr Johnson – Dictionary writer"  
coords="133,343,124,287,159,224,189,228,195,291,222,  
311,209,343,209,354,243,362,292,466,250,463"  
shape="poly" href="/wiki/Samuel_Johnson">
```

```
<area title="Boswell – Biographer" alt="Boswell"  
coords="76,224,84,255,43,302,62,400,123,423,121,361,  
137,344,122,290,111,234,96,225"  
shape="poly" href="/wiki/James_Boswell">
```

```
<area title="Sir Joshua Reynolds – Host"  
alt="Sir Joshua Reynolds – Host"  
coords="190,276,208,240,229,228,247,238,250,258,286,  
319,282,323,223,323,220,301,200,295"  
shape="poly" href="/wiki/Joshua_Reynolds">
```

Imagemaps

- Add USEMAP attribute to tag to indicate a client-side imagemap, e.g.

```
<IMG SRC="images/banner.gif" USEMAP="#bannerbar">
```

- Different regions of the image are described using <MAP> tag, e.g.

```
<MAP NAME="bannerbar">
```

```
<AREA SHAPE="RECT" COORDS="10,10,50,50" HREF="p1.html">
```

```
<AREA SHAPE="RECT" COORDS="50,10,90,50" HREF="p2.html">
```

```
<AREA SHAPE="RECT" COORDS="90,10,130,50" HREF="p3.html">
```

```
<AREA NOHREF SHAPE=default>
```

```
</MAP>
```

- Possible values for SHAPE are:
 - default**: Specifies the entire region.
 - rect**: Defines a rectangular region.
 - circle**: Defines a circular region.
 - poly**: Defines a polygonal region.

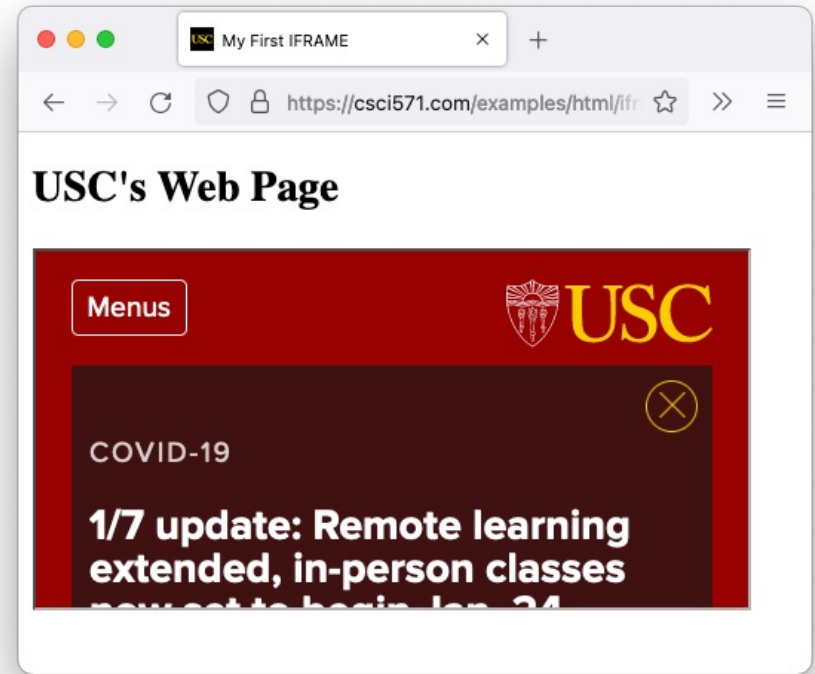
HTML Inline Frame

- Used to “embed” a website

```
<!DOCTYPE html>
<html>
<head>
  <title>My First IFRAME</title>
</head>
<body>
  <h2>USC's Web Page</h2>
```

```
  <iframe src="https://www.usc.edu" height="200" width="400" />
```

```
</body>
</html>
```



<META> Element

- Allows you to insert Name/Value pairs describing document properties, e.g.

```
<META NAME="Author" CONTENT="Ellis Horowitz">
```

- USC CS dept home page header

```
<META name="description"
```

```
content="The Computer Science Department at the University  
of Southern California, Los Angeles (USC) provides  
education leading to the Bachelors, Masters and Ph.D.  
degrees in Computer Science.">
```

```
<META name="keywords" content="USC, computer science,  
computer science research, computer science teaching">
```

```
<META name="author" content="Ellis Horowitz">
```

<META> Element

- Moving a Web page to a new site

```
<HTML>
```

```
<HEAD>
```

```
  <META HTTP-EQUIV="REFRESH" CONTENT="5; URL=http://www.usc.edu/dept/cs/">
```

```
  <META NAME="GENERATOR" CONTENT="Mozilla/4.04 [en] (Win95; I) [Netscape]">
```

```
  <TITLE>This site has moved</TITLE>
```

```
</HEAD>
```

```
<BODY>
```

```
  <CENTER>This site has moved to a new location which is:
```

```
    <A HREF="http://www.usc.edu/dept/cs/"> http://www.usc.edu/dept/cs/</A><BR>
```

```
Your browser should automatically move to the correct URL in five seconds.
```

```
  </CENTER>
```

```
</BODY>
```

```
</HTML>
```

Meta Tag and Robot Exclusion

```
<meta name="robots" content="noindex,nofollow">
```

```
<title>...</title> </head> <body> ...
```

- The content of the Robots META tag contains directives separated by commas.
- The currently defined directives are
 - **[NO] INDEX.** The INDEX directive specifies if an indexing robot should index the page.
 - **[NO] FOLLOW.** The FOLLOW directive specifies if a robot is to follow links on the page.
 - The defaults are INDEX and FOLLOW. The values ALL and NONE set all directives on or off: ALL=INDEX,FOLLOW and NONE=NOINDEX,NOFOLLOW.
- Some examples:

```
<meta name="robots" content="index, follow">
```

```
<meta name="robots" content="noindex, follow">
```

```
<meta name="robots" content="index, nofollow">
```

```
<meta name="robots" content="noindex, nofollow"> Note the  
"robots" name of the tag and the content are case  
insensitive.
```

Validating Your HTML

- The reasons for validation
 - Browsers display HTML differently
 - Browsers treat HTML errors differently
- What validators do
 - Flag syntax errors with respect to HTML DTD
 - Compare your pages to HTML 4.x, XHTML, and even HTML 5 (experimental)
- Some tools are downloaded to your site; others read your Web page from a URL
- HTML validation tools can be found at:
<http://search.yahoo.com/bin/search?p=html+validation>
- W3C Markup Validation Service:
<http://validator.w3.org/>

W3C Markup Validation Service

- Options: character encoding & Document type



The screenshot shows the W3C Markup Validation Service interface in a web browser. The browser's address bar displays `validator.w3.org/#validate_by_uri+with_options`. The page has a blue header with the W3C logo and the text "Markup Validation Service" and "Check the markup (HTML, XHTML, ...) of Web documents". Below the header, there are three tabs: "Validate by URI", "Validate by File Upload", and "Validate by Direct Input". The "Validate by URI" tab is selected. Under this tab, there is a section "Validate by URI" with the text "Validate a document online:". Below this, there is a text input field labeled "Address:" containing the URL `http://www.usc.edu`. Below the input field, there is a section "More Options" with several settings: "Character Encoding" set to "(detect automatically)" with a dropdown arrow and a checkbox "Only if missing"; "Document Type" set to "(detect automatically)" with a dropdown arrow and a checkbox "Only if missing"; radio buttons for "List Messages Sequentially" (selected) and "Group Error Messages by Type"; checkboxes for "Show Source", "Clean up Markup with HTML-Tidy", "Show Outline", "Validate error pages", and "Verbose Output". A "Check" button is located below these options. Below the form, there is a paragraph of text explaining the validator's purpose and providing links to other resources. At the bottom, there is a Mozilla logo and text stating that the W3C validators are developed with assistance from the Mozilla Foundation. There is also a "Donate" button and a "Flattr" button. The footer contains navigation links (Home, About..., News, Docs, Help & FAQ, Feedback, Contribute) and copyright information.

W3C® Markup Validation Service
Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI Validate by File Upload Validate by Direct Input

Validate by URI
Validate a document online:

Address:

▼ More Options

Character Encoding: (detect automatically) ☐ Only if missing

Document Type: (detect automatically) ☐ Only if missing

☒ List Messages Sequentially ☐ Group Error Messages by Type

☐ Show Source ☐ Clean up Markup with HTML-Tidy

☐ Show Outline ☐ Validate error pages ☐ Verbose Output

This validator checks the [markup validity](#) of Web documents in HTML, XHTML, SMIL, MathML, etc. If you wish to validate specific content such as [RSS/Atom feeds](#) or [CSS stylesheets](#), [MobileOK content](#), or to [find broken links](#), there are [other validators and tools](#) available.

 The W3C validators are developed with assistance from the Mozilla Foundation, and supported by community donations.

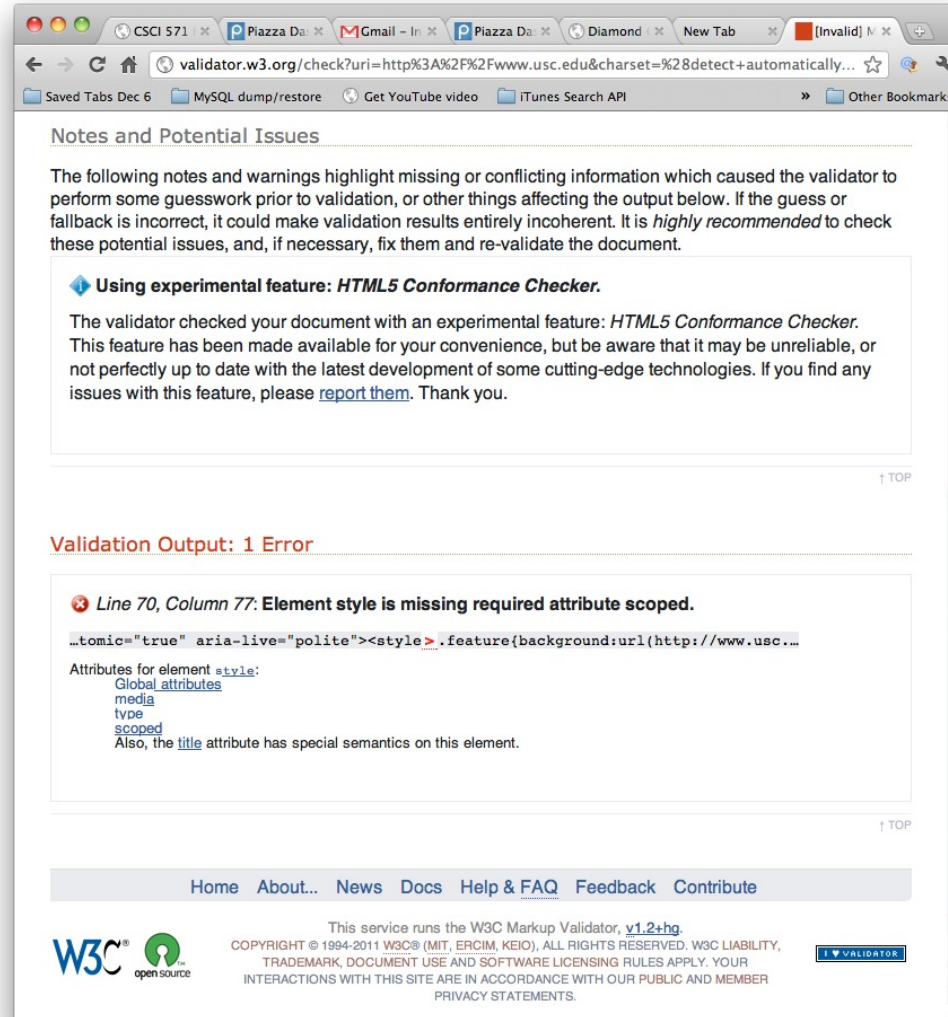
[Donate](#) and help us build better tools for a better web.

Home About... News Docs Help & FAQ Feedback Contribute

W3C®  This service runs the W3C Markup Validator, v1.2
COPYRIGHT © 1994-2010 W3C® (MIT, ERCIM, KEIO). ALL RIGHTS RESERVED. W3C LIABILITY, TRADEMARK, DOCUMENT USE AND SOFTWARE LICENSING RULES APPLY. YOUR INTERACTIONS WITH THIS SITE ARE IN ACCORDANCE WITH OUR PUBLIC AND MEMBER PRIVACY STATEMENTS.

Validating Your HTML

- Sample output



Helpful Links to Play with

- W3 Schools
<https://www.w3schools.com/html/>
- MDN (Mozilla Developer Network)
<https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- MDN Recommended articles
<https://developer.mozilla.org/en-US/curriculum/>
- CODEPEN
<https://codepen.io/>
- JS Fiddle
<https://jsfiddle.net/>
- JS BIN
<https://jsbin.com/?html,output>